

SPICSCON 2026 Full Step-by-Step Execution Plan

From topic selection and literature gap to simulation, results, IEEE paper writing, and submission readiness

Working paper direction: Topology-Aware Adaptive Fusion of WiFi RSSI and Visible-Light Signals for Robust Simulated Indoor Positioning

Core safe claim: The framework improves robustness in simulated hybrid WiFi-visible-light indoor localization under controlled noisy and blocked signal conditions. It does not claim real-world deployment, state-of-the-art performance, a new WiFi/VLP model, or a new RL algorithm.

Item	Decision
Paper type	Focused IEEE-style 4-6 page conference paper.
Evidence style	Simulation-based, physically motivated signal models, repeated seeds, honest ablations.
Must not become	A Q1-level hardware paper, a new VLP hardware paper, or an end-to-end RL localization paper.
Main contribution	A compact reproducible stress-test combining DSP denoising, graph smoothing, and adaptive fusion.
Most urgent next action	Create project skeleton, collect must-cite PDFs/Poompat slide evidence, and implement environment + signal models first.

Prepared for project resource upload and step-by-step execution. Generated on May 27, 2026.

0. How to Use This Plan

This document is a practical execution manual. Use it as the single roadmap inside the project resources. The plan is intentionally strict because the conference deadline is short and the paper must remain honest and evidence-driven.

- **Read Sections 1-3 first** to lock the paper scope, title, research questions, and contribution boundaries.
- **Use Sections 4-11 as the coding order.** Do not jump to RL before environment, signal generation, baselines, DSP, and GSP are working.
- **Use Sections 12-14 for result generation.** Every plot/table must be saved into the repo before writing claims.
- **Use Sections 15-18 to write the IEEE paper.** The writing must follow results, not imagination.
- **Use Section 19 as the rescue plan.** If adaptive fusion is unstable, the paper can still survive as DSP+GSP adaptive heuristic fusion.

Golden rule: If a result is not saved as a figure/table/log in the repo, do not claim it in the paper.

Dependency Chain

Order	Stage	You may move forward only when...
1	Scope and claim lock	Title, RQs, baselines, metrics, and claim boundary are fixed.
2	Literature evidence	10-12 key papers are downloaded or saved, and each has method-limit-gap notes.
3	Repo skeleton	All folders and config files exist; code can be run from one command.
4	Environment + trajectories	A layout plot and at least one trajectory plot are generated.
5	Signal models	WiFi and VLP signals are plotted under clean/noisy/blockage cases.
6	Baselines	WiFi-only, VLP-only, and static fusion produce errors and results CSV.
7	DSP + GSP	Filtering and graph smoothing generate ablation results and heatmaps.
8	Adaptive fusion	Bandit/Q-learning or heuristic adaptive fusion beats at least one fixed fusion in mixed scenario.
9	Final experiment run	All baselines run across seeds and scenarios; result tables are saved.
10	Paper writing	Paper text cites exact figures/tables and uses only supported claims.

1. Final Paper Target and Scope Lock

Before coding, lock the exact paper identity. This protects the work from becoming too broad and prevents overclaiming.

Final Recommended Title

Topology-Aware Adaptive Fusion of WiFi RSSI and Visible-Light Signals for Robust Simulated Indoor Positioning

This title is safer than the original Graph-DSP-RL title because it keeps the WiFi-visible-light fusion idea central, includes topology-awareness, avoids implying a new RL algorithm, and clearly signals simulation.

Core Thesis Sentence

A controlled simulation can test whether simple DSP denoising, graph-topology smoothing, and lightweight adaptive fusion improve robustness of hybrid WiFi RSSI-like and visible-light intensity-like localization when WiFi noise/multipath and visible-light blockage occur.

Must-have Scope

- 10 m x 10 m indoor room represented as a 10 x 10 grid graph with 100 candidate location cells.
- 2 or 3 WiFi access points and 4 LED/light anchors with fixed known coordinates.
- Synthetic walking trajectories using random waypoints or smooth grid-constrained paths.
- WiFi RSSI-like signals using log-distance path loss with Gaussian noise, multipath-like disturbances, and body-shadowing-like attenuation.
- Visible-light intensity-like signals using inverse-square attenuation with noise and blockage loss.
- Four scenarios: clean, WiFi degraded, VLP blocked, and mixed dynamic degradation.
- Baselines: WiFi-only, VLP-only, static fusion, DSP fusion, DSP+GSP, adaptive fusion, full proposed method.
- Metrics: mean error, median error, 90th percentile error, cell/zone accuracy, failure rate, recovery time, trajectory smoothness, runtime.

Explicitly Out of Scope

- Real-world deployment claims.
- State-of-the-art comparison against all indoor localization systems.
- A new WiFi propagation model.
- A new VLP optical channel model.
- A new reinforcement-learning algorithm.
- Camera/photodiode hardware validation unless done as a clearly optional qualitative demo.
- QFlow or quantum routing integration in this conference paper.

2. Final Research Questions, Hypotheses, and Evidence

RQ	Research Question	Hypothesis	Evidence Needed
RQ1	To what extent do simple DSP filters improve localization robustness for simulated WiFi RSSI-like and visible-light intensity-like signals under noisy indoor conditions?	EMA, median, and moving average filters reduce signal variance and improve localization compared with raw signals, especially under WiFi noise.	Noisy-vs-filtered signal plot; filter comparison table; mean/median/P90 error and runtime.
RQ2	Does graph Laplacian smoothing over an indoor grid improve spatial consistency and reduce localization error compared with non-topology-aware fusion?	Graph smoothing reduces isolated probability spikes and impossible trajectory jumps, especially in mixed degradation.	Raw vs graph-smoothed heatmap; trajectory smoothness; failure rate; ablation table.
RQ3	Can a lightweight adaptive fusion policy outperform fixed fusion when WiFi and visible-light reliability changes over time?	A contextual bandit or Q-learning policy selects better fusion modes than static weights under changing reliability.	Static vs adaptive fusion table; scenario-wise error plot; reward/mode-selection curve if stable; recovery metrics.

Keep exactly three RQs. More RQs will make the conference paper look over-expanded and will weaken the execution.

3. Literature and Gap Work: What to Read, Extract, and Write

The aim is not to read endlessly. The aim is to extract enough evidence to justify the narrow gap, support the baselines, and write a concise related-work section.

Must-Cite Paper Groups

Group	Papers / IDs	Use in Paper
WiFi indoor positioning	RADAR, Horus, WiDeep, DLoc, WiFi IPS surveys	Motivate RSSI/RF indoor positioning and noise/multipath limitations.
Visible-light positioning	Do and Yoo survey; Luo survey; Rahman survey; Afzalan real-world VLP survey	Motivate VLP modality and its line-of-sight, geometry, receiver, and blockage limitations.
Hybrid RF-optical / WiFi-VLP	Albraheem 2023, Lu 2025 RF-VLP, Mustafa and Hamza 2025 LiFi/WiFi	Show that hybrid RF-optical localization exists; narrow your gap to reproducible topology-aware adaptive stress-test.
Graph / GSP	Shuman 2013, Ortega 2018, graph-based indoor localization papers	Justify graph Laplacian smoothing over a grid/probability map.
RL/adaptive fusion	RL-based information fusion/localization papers	Justify lightweight adaptive fusion but avoid claiming a new RL algorithm.
Poompat VLP line	Saengudomlert EKF VLP, dimmable VLP, interference mitigation, slide photos	Use as visible-light inspiration and practical VLP background, not as your direct method.

Paper Note Template

For each of the 10-12 must-cite papers, write exactly these bullets in **notes/literature_notes.md**:

Paper ID / citation key:

1. Method: What signal modality and algorithm did the paper use?
2. Evidence: Was it simulation, dataset, or hardware? What metric did it report?
3. Limitation/gap: What does it not cover that your paper can honestly address?
4. How to cite: Which sentence in your paper will cite it?

Related Work Paragraph Plan

Paragraph	Main Message	Connects to Your Method
P1: Indoor positioning and WiFi	WiFi RSSI/fingerprinting is mature and low-cost, but signal instability and multipath remain hard.	WiFi RSSI-like model and WiFi-only baseline.
P2: VLP	Visible-light positioning can be accurate but depends on geometry, LOS, lighting, receiver conditions, and blockage.	VLP intensity-like model and blockage scenario.
P3: Hybrid RF-optical	RF and optical modalities fail differently; hybrid fusion is promising, but fixed/static fusion can be weak under changing reliability.	Static fusion baseline and adaptive fusion.
P4: DSP/GSP/adaptive fusion	Filtering, graph smoothing, and adaptive/RL fusion are known components, but a compact reproducible stress-test for WiFi-VLP under controlled failure is useful.	Your exact contribution and gap.

4. Project Folder Setup

Create the project as a reproducible research repository. Every generated result must be traceable to a config, seed, and script.

Folder Structure

```
spicscon2026-graph-dsp-rl-indoor/
```

```
README.md
```

```
requirements.txt
```

```
configs/
```

```
default.yaml
```

```
ablation.yaml
```

```
src/
```

```
environment.py
```

```
signal_models.py
```

```
dsp_filters.py
```

```
probability_maps.py
```

```
gsp_smoothing.py
```

```
rl_fusion.py
```

```
baselines.py
```

```
metrics.py
```

```
run_experiments.py
```

```
plot_results.py
```

```
utils.py
```

```
results/
```

```
raw/
```

```
processed/
```

```
figures/
```

```
tables/
```

```
logs/
```

```
notes/
```

```
literature_notes.md
```

```
poompat_slide_notes.md
```

```
experiment_diary.md
```

```
paper/
```

```
main.tex
```

```
references.bib
```

```
figures/
```

```
tables/
```

```
docs/
```

```
plan/
```

```
progress/
```

First Commands

```
mkdir -p spicscon2026-graph-dsp-rl-indoor/{configs,src,results/{raw,processed,figures,tables,logs},notes,paper/{figures,tables},docs/{plan,progress}}

cd spicscon2026-graph-dsp-rl-indoor

python3 -m venv .venv

source .venv/bin/activate

pip install numpy scipy pandas matplotlib scikit-learn networkx pyyaml tqdm seaborn

pip freeze > requirements.txt

git init
```

Note: You can use Matplotlib normally for project plots. The no-seaborn restriction only applies when I generate charts directly for you inside this chat.

5. Configuration File First

The configuration file should be created before coding so that experiments are reproducible. Do not hard-code anchor coordinates or noise levels inside scripts.

Minimum default.yaml

```
project:
name: spicscon2026_graph_dsp_rl_indoor
seed_train: [1,2,3,4,5]
seed_test: [101,102,103,104,105]
room:
width_m: 10
height_m: 10
grid_size: 10
anchors:
wifi_ap: [[1,1], [9,1], [5,9]]
leds: [[2,2], [8,2], [2,8], [8,8]]
trajectory:
num_trajectories: 50
steps_per_trajectory: 100
max_step_m: 0.5
wifi_model:
p0_dbm: -40
path_loss_n: 2.2
base_sigma_db: 2.0
noisy_sigma_db: 6.0
multipath_burst_prob: 0.12
shadowing_db_range: [3, 9]
vlp_model:
led_power: 1.0
receiver_height_m: 1.0
ceiling_height_m: 3.0
base_sigma: 0.02
blockage_loss: 0.75
blockage_prob: 0.20
dsp:
moving_average_window: 5
median_window: 5
ema_alpha: 0.35
gsp:
neighbor_type: 4
smoothing_alpha: 0.25
lambda_values: [0.05, 0.1, 0.2, 0.5]
metrics:
failure_threshold_m: 2.0
zone_radius_m: 1.0
```

6. Environment and Trajectory Implementation

Start coding from the environment. If the room, grid, anchors, and trajectories are wrong, every later result becomes meaningless.

File	Functions	Output to Save	Pass Condition
src/environment.py	create_grid(), cell_centers(), make_graph(), create_anchors(), generate_trajectory(), generate_scenarios()	results/figures/fig1_layout_trajectory.png; results/raw/trajectories_seed101.csv	You can plot the 10x10 grid, APs, LEDs, and one trajectory.
src/utls.py	load_config(), set_seed(), save_json(), save_csv(), ensure_dirs()	results/logs/run_config.json	All scripts load same config and save outputs consistently.

Implementation Logic

- Each grid cell has an index from 0 to 99 and a center coordinate (x, y).
- A 4-neighbor grid graph connects up, down, left, and right cells. Use 8-neighbor only as optional sensitivity test.
- Trajectories should be smooth enough to look human-like. Avoid random teleportation because graph smoothing will look artificially strong.
- Create S1-S4 scenario flags per time step: clean, WiFi degraded, VLP blocked, mixed dynamic.

Early Figure to Generate

Fig. 1: Room layout with WiFi APs, LED anchors, grid cells, and sample trajectory. This figure becomes the first method/setup figure in the paper.

7. Signal Model Implementation

Use standard physically motivated signal models. The goal is controlled stress testing, not accurate hardware-level channel modeling.

File	Model	Core Equation / Idea	Scenario Stress
src/signal_models.py	WiFi RSSI-like	$RSSI = P_0 - 10 n \log_{10}(d) + \text{noise} + \text{multipath/body-shadowing disturbance}$	S2 and S4 increase noise and add attenuation bursts.
src/signal_models.py	VLP intensity-like	$I = \text{LED_power} / (\text{distance}^2 + \text{epsilon}) + \text{noise} - \text{blockage_loss}$	S3 and S4 reduce intensity from one or more LEDs during blockage windows.

Pass Conditions

- For one trajectory, plot clean WiFi RSSI and degraded WiFi RSSI over time.
- For one trajectory, plot clean VLP intensity and blocked VLP intensity over time.
- Save raw signal arrays with seed and scenario names.
- Confirm that noise/blockage creates visible degradation but not impossible values everywhere.

Figures to Save

results/figures/fig3_noisy_vs_filtered_signals.png will later include raw and filtered signal examples. First generate the raw signal version, then add DSP outputs.

8. Probability Maps and Baselines

Before DSP, GSP, or adaptive fusion, you need a common probability/likelihood representation for every method.

Probability Map Concept

For each time step, compute a likelihood score for every grid cell. A cell receives high probability if the expected RSSI/intensity at that cell matches the measured signal. Normalize the 100-cell vector to sum to 1.

Baseline	How to Implement	Why It Is Required
WiFi-only	Generate WiFi likelihood over all cells; estimate cell with maximum probability or expected coordinate.	Shows WiFi weakness under RSSI fluctuation and multipath.
VLP-only	Generate VLP likelihood over all cells using LED intensity model.	Shows optical accuracy when clean and weakness under blockage.
Static fusion	$p_{\text{fused}} = w p_{\text{wifi}} + (1-w) p_{\text{vlp}}$ with $w = 0.5$ and optional WiFi-heavy/VLP-heavy variants.	Proves whether adaptive fusion adds value beyond simple weighted averaging.

Files

File	Functions
src/probability_maps.py	expected_wifi_at_cells(), expected_vlp_at_cells(), likelihood_gaussian(), normalize_prob(), estimate_position()
src/baselines.py	run_wifi_only(), run_vlp_only(), run_static_fusion(), run_weighted_fusion()
src/metrics.py	euclidean_error(), mean_error(), median_error(), p90_error(), zone_accuracy(), failure_rate()

Pass Condition

You can run WiFi-only, VLP-only, and static fusion for S1-S4 and produce a CSV table with mean, median, P90, and failure rate.

9. DSP Filtering Stage

DSP filtering is not the core novelty, but it is important because it creates a clear signal-processing contribution and supports RQ1.

Filter	Implementation	Expected Effect	Risk
Moving average	Average the last k samples per anchor signal.	Reduces random noise.	May lag trajectory changes.
Median filter	Median over sliding window per anchor signal.	Removes sudden spikes/outliers.	Can distort rapidly changing signal.
EMA	$y_t = \alpha x_t + (1-\alpha) y_{t-1}$	Smooth but responsive.	Alpha tuning can bias results.

Experiment for RQ1

- Run raw WiFi-only, raw VLP-only, and raw static fusion.
- Run the same methods after moving average, median, and EMA filtering.
- Report signal variance reduction and localization error changes.
- Pick one filter as the default for the full pipeline using validation seeds, not test seeds.

Output

Save results/tables/table_filter_comparison.csv and results/figures/fig3_noisy_vs_filtered_signals.png.

10. Graph Signal Processing / Graph Smoothing Stage

The graph stage is the main topology-aware part. Treat the 100-cell location probability vector as a graph signal over the indoor grid.

Two Safe Smoothing Options

Option	Formula / Idea	When to Use
Diffusion smoothing	$p_{\text{smooth}} = \text{normalize}((1-\alpha) p + \alpha A_{\text{norm}} p)$	Fast and simple. Good for conference implementation.
Laplacian regularized smoothing	Solve $(I + \lambda L) p_{\text{smooth}} = p$	More formal GSP presentation; use if implementation is stable.

Files and Outputs

File	Functions	Output
src/gsp_smoothing.py	build_grid_graph(), normalized_adjacency(), graph_diffusion(), laplacian_smoothing()	Graph object and smoothed probability maps.
src/plot_results.py	plot_probability_heatmap(), plot_raw_vs_smoothed_heatmap()	Fig. 4 raw vs graph-smoothed heatmap.

Experiment for RQ2

- Compare static fusion vs DSP fusion vs DSP+GSP.
- Report mean/median/P90 error, trajectory smoothness, and failure rate.
- Show one heatmap before and after graph smoothing at a noisy or blocked time step.
- Tune alpha/lambda only on validation seeds; freeze it for test seeds.

Warning

Do not say you introduce new graph theory. Say you apply a topology-aware graph smoothing layer to location probability maps.

11. Adaptive Fusion Stage

The adaptive fusion stage should remain lightweight. It should choose among predefined fusion modes; it should not learn localization from scratch.

Recommended Action Set

Action	Fusion Mode
A0	WiFi-heavy fusion: 0.75 WiFi + 0.25 VLP
A1	VLP-heavy fusion: 0.25 WiFi + 0.75 VLP
A2	Balanced fusion: 0.50 WiFi + 0.50 VLP
A3	Graph-prior-heavy fusion: apply stronger graph smoothing or combine fused likelihood with graph-smoothed probability
A4	Recovery mode: down-weight currently unreliable modality and rely on previous estimate + graph smoothing for short interval

State Features

- Recent WiFi variance over a sliding window.
- Recent VLP variance over a sliding window.
- WiFi likelihood entropy or max probability confidence.
- VLP likelihood entropy or max probability confidence.
- VLP blockage indicator from sudden intensity drop.
- Previous action or previous estimated jump size.

Policy Choice

Policy	Use When	Comment
Reliability heuristic	Very low time or RL unstable.	Still acceptable if called adaptive reliability-based fusion, not RL.
Contextual bandit	Recommended main option.	Chooses action based on current reliability features; simpler than Q-learning.
Tabular Q-learning	Use if state can be discretized cleanly.	Good if you want RL curve, but avoid complexity.
DQN	Only if all simpler methods are complete and stable.	Not recommended for 5-day paper.

Experiment for RQ3

- Train policy using training seeds under S1-S4.
- Evaluate only on held-out test seeds.
- Compare against fixed WiFi-heavy, VLP-heavy, balanced, graph-heavy, and static fusion.
- Report failure rate and recovery time in S3/S4 because adaptive fusion should shine there.
- Save action/mode timeline for one mixed trajectory to show behavior change during blockage/noise.

12. Metrics and Results Engine

The metrics script must be written before final experiments, so the result table is generated automatically and consistently.

Metric	Definition	Why It Matters
Mean localization error	Average Euclidean distance between true and estimated positions.	Main accuracy metric.
Median localization error	Median Euclidean error over time steps/seeds.	Robust to extreme failures.
90th percentile error	Error threshold below which 90 percent of samples fall.	Shows tail behavior under blockage/noise.
Cell/zone accuracy	Correct cell or within radius/zone of true position.	Easy interpretation for grid localization.
Failure rate	Percent of steps where error > fixed threshold, e.g., 2 m.	Robustness metric.
Recovery time	Steps needed to return below failure threshold after degradation event.	Supports self-healing/recovery claim.
Trajectory smoothness	Average estimated step jump or jerk-like change.	Shows graph smoothing reduces impossible jumps.
Runtime	Average computation time per time step or trajectory.	Supports low-resource feasibility.

Result CSV Files

```

results/tables/table_scenario_params.csv
results/tables/table_baseline_comparison.csv
results/tables/table_ablation.csv
results/tables/table_runtime.csv
results/raw/errors_by_seed_scenario_method.csv
results/logs/final_run_config.json
results/logs/git_commit.txt

```

Pass Condition

A single command should regenerate all final metrics from the same config: **python -m src.run_experiments --config configs/default.yaml --out results/**

13. Final Experiments and Ablation Order

Do not run the full expensive experiment before a small smoke test passes.

Run Level	Purpose	Command / Output	Stop Condition
Smoke test	Check environment, signal, baseline, metrics.	1 trajectory, 10 steps, 1 seed.	No crashes; figures and one result CSV saved.
Mini experiment	Check all baselines for S1-S4.	5 trajectories, 30 steps, 2 seeds.	Ranking makes sense; failure cases visible.
Validation run	Tune DSP alpha/window, GSP lambda, adaptive policy parameters.	Training/validation seeds only.	Choose final parameters and freeze config.
Final test run	Produce paper results.	Held-out test seeds only.	All tables/figures generated and saved.
Robustness run	Check whether results are seed-stable.	Additional seeds or sensitivity on noise/blockage.	At least direction of improvement is stable.

Ablation Ladder

WiFi-only -> VLP-only -> static fusion -> DSP fusion -> DSP+GSP -> adaptive fusion -> full Graph-DSP-adaptive method.

14. Figures and Tables for the IEEE Paper

For a 4-6 page IEEE paper, figures must be compact. Generate more plots for yourself, but include only the strongest ones in the paper.

Figure	Content	Supports	Priority
Fig. 1	Indoor layout with WiFi APs, LED anchors, and sample trajectory.	Simulation setup clarity.	Must-have
Fig. 2	Proposed architecture: signal models -> DSP -> probability maps -> GSP -> adaptive fusion.	Method overview.	Must-have
Fig. 3	Noisy vs filtered WiFi/VLP signal example.	RQ1.	Must-have
Fig. 4	Raw vs graph-smoothed probability heatmap.	RQ2.	Must-have
Fig. 5	Scenario-wise localization error across baselines.	Main quantitative result.	Must-have
Fig. 6	Adaptive fusion mode/reward curve.	RQ3, only if stable.	Conditional
Table	Content	Purpose	
Table I	Simulation/scenario parameters.	Reproducibility.	
Table II	Baseline comparison: mean, median, P90, failure rate.	Main quantitative result.	
Table III	Ablation: raw -> DSP -> DSP+GSP -> adaptive.	Shows component contribution.	
Table IV	Runtime/cost per step.	Supports lightweight feasibility.	

Space-Saving Rule

If limited to 4 pages, merge Fig. 3 and Fig. 4 into one two-panel figure, and merge Table II and Table III into one compact table.

15. IEEE Paper Writing Plan

Write the paper only after the first stable result table exists. Otherwise the paper becomes speculative.

Section	Target Length	What to Write	Evidence Needed
Abstract	150-200 words	Problem, method, simulation setting, best supported result, limitation.	Final Table II/III numbers.
Introduction	0.75 page	Indoor GPS problem, WiFi noise, VLP blockage, static fusion weakness, need for topology-aware adaptive fusion.	WiFi/VLP/hybrid citations and strongest gap sentence.
Related Work	0.6-0.8 page	WiFi, VLP, hybrid RF-optical, graph/GSP, adaptive fusion in concise paragraphs.	10-12 citations only.
System Model	0.6 page	Room, anchors, trajectory, WiFi equation, VLP equation, scenarios.	Config and Table I.
Proposed Framework	0.9 page	DSP filters, probability maps, GSP smoothing, adaptive fusion policy.	Fig. 2, equations, algorithm summary.
Experimental Setup	0.4 page	Baselines, metrics, seeds, implementation details.	Baseline list and metrics.
Results and Discussion	1.3-1.6 pages	Answer RQ1-RQ3 using figures/tables. Explain not just numbers but why failures happen.	Figs. 3-6, Tables II-IV.
Limitations	0.25 page	Simulation-only, simplified channel models, no hardware validation yet.	Claim boundary.
Conclusion	0.2 page	One short summary and future hardware validation.	No new claims.

Recommended Page Budget

Target 5 pages if allowed: Page 1 = title/abstract/intro; Page 2 = related work + system model; Page 3 = method; Page 4 = experiment setup + first results; Page 5 = main results, limitations, conclusion, references. If 6 pages allowed, use the sixth page for clearer results and references.

16. Section-by-Section Writing Checklist

Use this checklist while drafting main.tex.

Section	Must Include	Must Avoid
Abstract	Simulation, hybrid WiFi-VLP, DSP, graph smoothing, adaptive fusion, key metric.	SOTA, deployment, hardware claims.
Introduction	Problem, limitations of single modalities, gap, contributions.	Too much generic GPS text or long history.
Related Work	Only the papers that directly frame the gap.	Long literature survey that consumes space.
System Model	Equations and parameter table.	Unexplained constants or hidden assumptions.
Method	Clear pipeline and algorithmic steps.	Making RL look bigger than it is.
Experiments	Baselines, scenarios, seeds, metrics.	Unfair tuning on test seeds.
Results	RQ-by-RQ analysis, not just tables.	Claiming improvements without discussing failure cases.
Limitations	Simulation-only and simplified modeling.	Hiding weaknesses.
Conclusion	Repeat supported contribution and future validation.	Introducing new ideas/results.

Contribution Wording to Use

- A reproducible simulation framework for hybrid WiFi RSSI-like and visible-light intensity-like indoor positioning under controlled clean, WiFi-degraded, VLP-blocked, and mixed-degradation scenarios.
- A topology-aware graph smoothing layer that treats grid-cell location probabilities as graph signals and reduces noisy probability spikes and impossible trajectory jumps.
- A lightweight adaptive fusion policy that selects among predefined WiFi-heavy, VLP-heavy, balanced, graph-prior, and recovery modes, evaluated against fixed-fusion and ablation baselines.

Gap Sentence to Use

Existing WiFi, VLP, graph-based, and adaptive-fusion works often study individual components or hardware-specific systems. This paper addresses the narrower need for a reproducible simulation study that isolates how DSP denoising, graph-topology smoothing, and lightweight adaptive fusion behave under controlled WiFi noise/multipath and visible-light blockage.

17. Five-Day Sprint Plan

Day	Main Target	Concrete Output	Stop Condition
Day 1	Finalize scope, citation evidence, and repo config.	notes/literature_notes.md, references.bib, configs/default.yaml, paper outline.	You can explain the gap, RQs, baselines, metrics, and claim boundary in 2 minutes.
Day 2	Implement environment and signal models.	Layout figure, trajectories, clean/noisy/blockage signal plots, raw signal CSVs.	You can plot S1-S4 signals for one trajectory and show degradation works.
Day 3	Implement baselines, DSP, and GSP.	Initial baseline table, filter comparison, heatmaps, ablation CSV.	WiFi-only, VLP-only, static, DSP, and GSP all run across scenarios.
Day 4	Implement adaptive fusion and final experiments.	Final baseline table, ablation table, runtime table, figures.	All main metrics run over repeated held-out seeds and are saved to results/.
Day 5	Write, polish, compile, and check claims.	main.tex, references.bib, final PDF, response-to-checklist.	Paper compiles, all figures/tables cited, no unsupported claim remains.

Daily Work Rhythm

- Morning: implement or write the next missing block.
- Afternoon: run experiments or generate figures/tables.
- Night: write the paper section that corresponds to the evidence produced that day.
- End of each day: save outputs under results/ or docs/ and commit to Git.

18. What to Find Next

Do not search randomly. Find only the items below because they directly affect execution and writing.

Priority	What to Find	Why It Matters	Where to Save
1	SPICSCON official paper template, page limit, file size, deadline/time zone, author guidelines.	Prevents formatting/submission rejection.	notes/conference_rules.md and paper/template/.
2	Poompat slide photos.	Identifies exact VLP presentation and whether it maps to EKF VLP, dimmable VLP, or wheelchair tracking.	notes/poompat_slide_notes.md and evidence_to_verify/poompat_slides/.
3	PDFs or official pages for 10-12 must-cite papers.	Needed for related work and reference accuracy.	evidence_to_verify/papers/.
4	Exact DOI/venue for uncertain Poompat papers P40/P41 if you plan to cite them.	Avoids unverified citations.	notes/literature_notes.md.
5	Final simulation parameter values.	Needed before coding final config.	configs/default.yaml.
6	IEEE LaTeX template and bibliography style.	Needed for final paper.	paper/.

Must-Collect Citation Set

P21, P22, P36, P38, P20, P27, P25, P23/P24, P29, P05/P06, P16/P15. These cover hybrid RF-VLP, LiFi/WiFi, Poompat VLP, graph localization/GSP, adaptive fusion, WiFi robustness, and VLP surveys.

Most Important Practical Question

Can the full method consistently improve failure rate and recovery time in S3/S4 compared with static fusion? If yes, the paper has a strong conference story. If not, make the paper a transparent ablation study and reduce the adaptive fusion claim.

19. Rescue Plans

Problem	Rescue Action	How to Reframe Paper
Adaptive fusion/RL does not learn well.	Use reliability-based adaptive heuristic and call it lightweight adaptive fusion, not RL.	Focus on DSP+GSP and reliability-aware mode switching.
Graph smoothing hurts clean-case accuracy.	Report that it improves noisy/mixed robustness but may oversmooth clean cases; tune lambda on validation seeds.	This is a useful honest discussion.
VLP model gives unrealistic high accuracy.	Increase noise/blockage, add receiver-height distance, or report relative robustness rather than absolute accuracy.	Emphasize controlled stress-test, not real VLP accuracy.
No time for optional demo.	Drop it completely.	Simulation-only is acceptable if claims are honest.
Too many figures/tables for 4 pages.	Merge figures and tables; keep only setup, architecture, main result, ablation.	Prioritize evidence over decoration.

20. Final Claim Boundary Table

Safe Claim	Avoid This Claim
Improves robustness in simulated noisy and blocked signal conditions.	Solves indoor positioning.
Uses standard physically motivated WiFi RSSI-like and VLP intensity-like models.	Proposes a new WiFi/VLP signal model.
Evaluates lightweight adaptive fusion for selecting predefined modes.	Introduces a new RL algorithm or end-to-end RL localization.
Graph smoothing improves spatial consistency in the configured simulation.	Introduces new GSP theory or SOTA graph localization.
Provides a reproducible open-source simulation testbed.	Provides deployment-ready real-world localization.
Outperforms fixed-fusion baselines under tested simulated scenarios.	Achieves state-of-the-art performance against all indoor localization systems.

21. Final Pre-Submission Checklist

Use this before submitting or sending the paper to a supervisor.

Check	Question
Scope	Does the paper clearly say simulation-based and physically motivated models?
Gap	Does the introduction state a narrow reproducible stress-test gap, not a broad novelty claim?
Related work	Are hybrid RF/VLP, LiFi/WiFi, Poompat VLP, GSP, and adaptive fusion papers cited?
Config	Are room size, anchors, scenarios, seeds, and thresholds listed?
Baselines	Are WiFi-only, VLP-only, static fusion, DSP, GSP, and adaptive/full methods included?
Metrics	Are mean, median, P90, failure rate, recovery time, smoothness, and runtime reported?
Ablation	Does the paper show each component contribution?
Figures	Are all figures readable in IEEE two-column format?
Tables	Are values rounded consistently and not overloaded?
Claims	Are all unsupported SOTA/deployment/new-model/RL-novelty claims removed?
Reproducibility	Can one command regenerate the main result tables?
Submission	Does PDF satisfy page limit, file size, template, and deadline rules?

Final Go/No-Go Rule

Go if the full method improves at least two of these under S3/S4: P90 error, failure rate, recovery time, or trajectory smoothness. If improvement appears only in mean error, the paper is weaker; focus the discussion on ablation and limitations.

22. One-Command Target Workflow

The final project should ideally run with commands like the following:

```
# 1. Create layout and signal sanity plots
python -m src.environment --config configs/default.yaml --plot
python -m src.signal_models --config configs/default.yaml --plot

# 2. Run smoke test
python -m src.run_experiments --config configs/default.yaml --mode smoke

# 3. Run validation/tuning
python -m src.run_experiments --config configs/default.yaml --mode validation

# 4. Run final test experiments
python -m src.run_experiments --config configs/default.yaml --mode final

# 5. Generate all figures/tables
python -m src.plot_results --config configs/default.yaml --input results/raw --out results

# 6. Copy final figures/tables to paper directory
cp results/figures/*.png paper/figures/
cp results/tables/*.csv paper/tables/

# 7. Compile paper
cd paper
pdflatex main.tex
bibtex main
pdflatex main.tex
pdflatex main.tex
```

23. Copyable Instruction for Future Project Chats

Paste this into future coding/writing chats to keep them aligned:

I am working on an IEEE-style SPICSCON 2026 conference paper titled "Topology-Aware Adaptive Fusion of WiFi RSSI and Visible-Light Signals for Robust Simulated Indoor Positioning." The paper is simulation-based and must not overclaim. Build a reproducible Python project that models a 10 m x 10 m indoor grid, 3 WiFi APs, 4 LED anchors, synthetic walking trajectories, WiFi RSSI-like path-loss signals, VLP intensity-like signals, DSP filters, graph Laplacian smoothing, and lightweight adaptive fusion. Compare WiFi-only, VLP-only, static fusion, DSP fusion, DSP+GSP, adaptive fusion, and full method across clean, WiFi-noisy, VLP-blocked, and mixed dynamic scenarios. Report mean/median/P90 error, zone accuracy, failure rate, recovery time, trajectory smoothness, and runtime. Save every result under results/ and every paper figure/table under paper/. Claims must stay simulation-only and evidence-based.

End of execution plan.